EduAssist AI: API Cost, Latency and Prompt Version Management

MLOps (AIMLCZG523), Experiential Learning Assignment 2

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1. Setup

- LLMOps builds on MLOps. Prompts and model versions are versioned, and cost, latency and tokens are tracked for each request.
- The use case is EduAssist AI, an assistant that answers academic questions.
- All results below come from one run on the BITS MLOps lab VM.

Table 1. Run details

Item	Details
Model	gemini-3.5-flash-lite
API version	Gemini API v1beta (google-genai 2.34.0)
Settings	Temperature 0, maximum 2,048 output tokens
Pricing	$0.30 input and $2.50 output per million tokens (USD)
Pricing date	30 October 2026
Billing	Free tier was used, so costs shown are at paid-tier rates
Lab environment	BITS MLOps lab VM, Python 3.12.9, MLflow 3.17.2
API key	Read from an environment variable; a scan found it in no output file

2. Task 1: Latency

Table 2. Latency for five questions (prompt v1, all requests successful)

ID	Question topic	Latency (s)	Input tokens	Output tokens
Q1	Overfitting and prevention	4.448	25	680
Q2	TCP versus UDP	3.645	20	566
Q3	Photosynthesis	2.124	15	208
Q4	Binary search complexity	3.957	23	547
Q5	2008 financial crisis	3.825	28	844

Latency summary (seconds): mean 3.600, median 3.825, minimum 2.124, maximum 4.448, standard deviation 0.877.

Why Q1 was the slowest (4.45 s):

- Longer answers generally took longer (r = 0.83), because the model writes one token at a time.
- Length is not the full reason. Q5 gave more tokens than Q1 but finished sooner.
- Q1 was the first call of the run, so connection set-up probably added time.
- Server load and network delay may also have played a part.

3. Task 2: Tokens and Cost

Table 3. Tokens and cost per request (prompt v1)

ID	Input tokens	Output tokens	Total tokens	Cost (USD)
Q1	25	680	705	0.001708
Q2	20	566	586	0.001421
Q3	15	208	223	0.000525
Q4	23	547	570	0.001374
Q5	28	844	872	0.002118
Average	**22.2**	**569.0**	**591.2**	**0.001429**

Monthly projection: $0.001429 × 10,000 requests = $14.29.

Recommendation: shorten the answers.

- Output tokens make up over 99% of the cost. They are priced 8.3 times higher than input tokens, and answers averaged 569 tokens.
- A 250-token limit would bring the monthly cost down by 57%, to $6.11 (based on the measured tokens).
- Sending simple questions to gemini-3.1-flash-lite would save 40%, or 74% together with the limit.
- Cutting context or caching prompts would save very little, as input is a tiny part of the bill.

4. Task 3: Prompt Evaluation and Promotion

- Created the MLflow experiment `eduassist_llmops_assignment`.
- Registered v1, v2 and v3 as `eduassist-academic-qa`, each with a commit message.
- Loaded each prompt from the registry and ran it on the same five questions.
- Calculated `valid_json` as valid JSON responses divided by total responses.

Table 4. Prompt comparison

Version	Avg latency (s)	Avg input tokens	Avg output tokens	Avg cost (USD)	Valid JSON rate	Recommended?
v1	2.435	22.2	607.8	0.001526	0%	No
v2	1.931	46.2	376.4	0.000955	0%	No
v3	**1.323**	**87.2**	**150.8**	**0.000403**	**100%**	**Yes**

Promotion justification: v3 now has the `production` alias.

- **Selection rule:** a prompt must pass the JSON check first. Coverage, cost and latency are compared after that.
- **JSON validity:** the app reads every reply as JSON. v1 and v2 scored 0%, so they would break it. v3 scored 100% and matched the schema on all five questions.
- **Quality:** v3 covered 92% of expected concepts, while v1 and v2 covered 100%. We accept this gap because the app cannot use an answer it cannot read. v3 also gives key points and a confidence level.
- **Latency:** v3 averaged 1.32 s, which is 46% faster than v1.
- **Tokens:** v3 uses about four times the input of v1 but only a quarter of the output. Output is what drives time and cost, and the schema keeps answers short.
- **Cost:** v3 comes to $4.03 per 10,000 requests, against $15.26 for v1 and $9.55 for v2. That is 74% cheaper than v1. The extra input tokens are cheap, so they add very little.

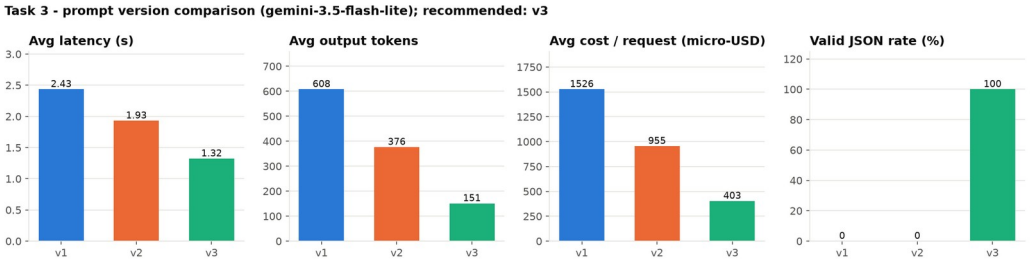


Figure 1. Prompt comparison from the promotion_decision run in MLflow.

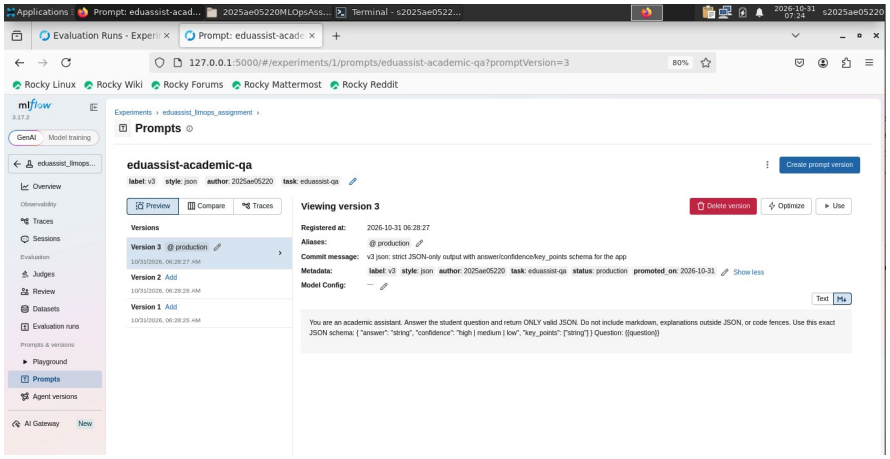


Figure 2. Prompt Registry showing version 3 with the production alias.

5. Limitations

- Five questions and one run is a small sample.
- Results can change between runs. Prompt v1 ran twice: 3.60 s and $14.29 in Task 1, then 2.43 s and $15.26 in Task 3.
- Keyword coverage is only a rough measure of quality.
- Next steps: use more questions, add human review, and monitor valid_json in production.
- The app loads the prompt by alias, so rollback needs just one alias change.

6. Evidence

All files are in the submission ZIP and at github.com/gsivaram/2025ae05220MLOpsAssignment2.

Table 5. Evidence by section

Section	File name
Setup and reproduction	README.md requirements.txt
Lab run	screenshots/lab_00_environment.png screenshots/lab_01_pipeline_passed.png reports/run/run_summary.md
All tasks (code and outputs)	notebooks/EduAssist_LLMOps_Assignment2.ipynb reports/EduAssist_LLMOps_Assignment2.html
Task 1: Latency	outputs/latency_tokens_cost.csv outputs/latency_summary.csv outputs/figures/task1_latency.png
Task 2: Tokens and cost	outputs/latency_tokens_cost.csv outputs/figures/task2_cost_per_request.png

Section	File name
Task 3: Runs and metrics	screenshots/mlflow_01_experiment_runs.png screenshots/mlflow_02a_metrics_compare_v1_v2.png screenshots/mlflow_02b_metrics_compare_v2_v3.png screenshots/mlflow_03a_run_overview.png screenshots/mlflow_03b_run_overview.png
Task 3: Evaluation	screenshots/mlflow_04_evaluation.png outputs/prompt_evaluation_results.csv outputs/prompt_evaluation_details.csv
Task 3: Prompt Registry	screenshots/mlflow_05_prompt_registry_v1.png screenshots/mlflow_05_prompt_registry_v2.png screenshots/mlflow_05_prompt_registry_v3.png
Task 3: Prompt diff	screenshots/mlflow_06a_prompt_diff.png screenshots/mlflow_06b_prompt_diff.png outputs/prompt_diffs.txt
Task 3: Production alias	screenshots/mlflow_07_production_alias.png
Task 3: Promotion decision	screenshots/mlflow_08_promotion_run.png outputs/promotion_justification.md